SDU 410: Indicates Open Circuit Fault – Low Speed Oil Pressure Switch

Symptoms

The fault message is displayed on the diesel control unit (DCU) 410E.

How To Use This Tree

Circuit Description

The SDU 410 has eight switch inputs. Each switch input has open circuit fault detection. The SDU 410 is monitoring the resistance of the circuit. The low speed oil pressure switch monitors oil pressure above 1400 RPM. A 10k ohm resistor is installed in the connector that mates to the switch.

Component Location

The SDU 410 is in the customer interface box.

Conditions for Running the Diagnostics

Customer interface box power switch ON.

Conditions for Setting the Code

The SDU 410 detects an open circuit. The overall resistance of the circuit is greater than 10k ohms.

Actions Taken when the Fault Code is Active

The DCU 410E will display one of the following faults:

Lube Oil Pressure Low (Low Speed)

Conditions for Clearing the Fault Code

SDU 410 detects adequate resistance on the affected circuit.

Acknowledge the fault on the DCU 410E.

Shoptalk

Possible causes include:

- Broken or disconnected wiring
- Damaged or missing open circuit detection resistor
- Malfunctioning switch

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check wiring connections.	
	STEP 1A. Check all wiring harness connection points.	Connections tight and secure?
STEP 2.	Check the low speed oil pressure switch.	
	STEP 2A. Check the low speed oil pressure switch.	Greater than 100k ohms?
	STEP 2B. Check the low speed oil pressure switch connector resistor.	Greater than 11k ohms?
STEP 3.	Check the low speed oil pressure switch wiring harness.	
	STEP 3A. Check the low speed oil pressure signal and return wires for an open circuit.	Greater than 10 ohms?
	STEP 3B. Check the low speed oil pressure signal and return wires for a wire-to-wire short.	Greater than 10 ohms?
	STEP 3C. Check the low speed oil pressure signal wire for a short to ground.	Greater than 10 ohms?

STEP 1. Check wiring connections.

STEP 1A. Check all wiring harness connection points.

Conditions :

- Engine OFF.
- Customer interface box power switch OFF.

Action	Specification/Repair	Next Step

Check the following connection points for secure connection. • SDU 410 terminal block connection inside the customer interface box. • Optional customer-provided circuit connections.	Connections tight and secure? **YES**	**2A**
	Connections tight and secure? **NO** **Repair :** Connect any disconnected harnesses. Repair or replace damaged connections. • Inside customer interface box: Refer to Procedure 015-138 in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-138.html) • Optional customer-provided circuit connections: See equipment manufacturer service information.	**Repair complete.**

STEP 2. Check the low speed oil pressure switch.

STEP 2A. Check the low speed oil pressure switch.

Conditions :

- Engine OFF.
- Customer interface box power switch OFF.

Action	Specification/Repair	Next Step

Disconnect the low speed lubricating oil pressure switch connector. Measure the resistance at the low speed lubricating oil pressure switch. • Place one test lead on the low speed lubricating oil pressure SIGNAL pin at the switch. • Place the other test lead on the low speed lubricating oil pressure RETURN pin at the switch. See the appropriate wiring diagram or pin and wire identification.	Greater than 100k ohms? **YES**	**2B**
	Greater than 100k ohms? **NO** **Repair :** Replace the low speed lubricating oil pressure switch. Refer to Procedure 015-141 in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-141.html)	**Repair complete.**

STEP 2B. Check the low speed oil pressure switch connector resistor.

Conditions :

- Engine OFF.
- Customer interface box power switch OFF.

Action	Specification/Repair	Next Step

Disconnect the low speed lubricating oil pressure switch connector. Measure the resistance across SUPPLY and the SIGNAL pins on the low speed oil pressure switch connector. • Place one test lead on the low speed lubricating oil pressure wiring harness SIGNAL pin. Place the other test lead on the SUPPLY pin. See the appropriate wiring diagram or pin and wire identification.	Greater than 11k ohms? **YES** **Repair :** Replace the resistor in the connector of the low speed oil pressure switch wiring harness.	**Repair complete.**
	Greater than 11k ohms? **NO**	**3A**

STEP 3. Check the low speed oil pressure switch wiring harness.

STEP 3A. Check the low speed oil pressure signal and return wires for an open circuit.

Conditions :

- Open the customer interface box.
- Disconnect the low speed oil pressure SIGNAL and RETURN wires at the SDU 410 unit and connector C4.

Action	Specification/Repair	Next Step

Check the low speed oil pressure SIGNAL and RETURN wires for an open circuit. • Place one test lead on the low speed oil pressure SIGNAL wire at the SDU 410 unit. • Place the other test lead on the low speed oil pressure SIGNAL pin at the C4 connector. • Place one test lead on the low speed oil pressure RETURN wire at the SDU 410 unit. • Place the other test lead on the low speed oil pressure RETURN pin at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Greater than 10 ohms? **YES**	**3B**
	Greater than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete.**

STEP 3B. Check the low speed oil pressure signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the low speed oil pressure SIGNAL and RETURN wires at the SDU 410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the low speed oil pressure SIGNAL and RETURN wires for a wire-to-wire short. • Place one test lead on the low speed oil pressure SIGNAL wire at the SDU 410 unit. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the low speed oil pressure RETURN wire at the SDU 410 unit. • Place the other test lead on all other pins at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Greater than 10 ohms? **YES**	**3C**
	Greater than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete.**

STEP 3C. Check the low speed oil pressure signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the low speed oil pressure SIGNAL wire at the SDU 410 unit and C4 connector.

Action	**Specification/Repair**	**Next Step**

Check the low speed oil pressure SIGNAL wire for a short to ground. - Place one test lead on the low speed oil pressure SIGNAL wire at the SDU 410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Greater than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete.**
	Greater than 10 ohms? **NO** **Repair :** Replace the SDU 410. Refer to Procedure 015-122 in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-122.html)	**Repair complete.**

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